



Original research

EMG of the hip adductor muscles in six clinical examination tests

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ABSTRACT

Objectives: To assess activation of muscles of hip adduction using EMG and force analysis during standard clinical tests, and compare athletes with and without a prior history of groin pain.

Study design: Controlled laboratory study.

Participants: 21 male athletes from an elite junior soccer program.

Main outcome measures: Bilateral surface EMG recordings of the adductor magnus, adductor longus, gracilis and pectineus as well as a unilateral fine-wire EMG of the pectineus were made during isometric holds in six clinical examination tests. A load cell was used to measure force data.

Results: Test type was a significant factor in the EMG output for all four muscles (all muscles $p < 0.01$). EMG activation was highest in Hips 0 or Hips 45 for adductor magnus, adductor longus and gracilis. EMG activation for pectineus was highest in Hips 90. Injury history was a significant factor in the EMG output for the adductor longus ($p < 0.05$), pectineus ($p < 0.01$) and gracilis ($p < 0.01$) but not adductor magnus. For force data, clinical test type was a significant factor ($p < 0.01$) with Hips 0 being significantly stronger than Hips 45, Hips 90 and Side lay. BMI (body mass index) was a significant factor ($p < 0.01$) for producing a higher force. All other factors had no significant effect on the force outputs.

Conclusions: Hip adduction strength assessment is best measured at hips 0 (which produced most force) or 45° flexion (which generally gave the highest EMG output). Muscle EMG varied significantly with clinical test position. Athletes with previous groin injury had a significant fall in some EMG outputs.

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1. Introduction

Over many years the treatment of groin pain in athletes has been difficult. Not only have the pathology of many conditions been ill-defined, so have assessment procedures (Fricker & Lovell, 2007). Athletes with or susceptible to groin pain are screened and assessed using 'hands on' clinical examination skills of which a series of resisted hip adduction tests play a considerable role. This role of hip adduction has been based on using inherently intuitive anatomically based clinical tests that have not been scientifically verified. Recently a number of authors have looked at the clinical examination of athletes with groin pain attempting to evaluate validity and reliability of particular tests (Fulcher, Hanna, & Raina Elley, 2010; Hogan & Lovell, 2002; Hölmich, Hölmich, & Bjerg, 2004; Malliaras, Hogan, Nawrocki, Crossley, & Schache, 2009). These tests of bilateral adduction of the legs against resistance have been

performed at various angles of hip flexion and knee flexion and are proving to be both valid and reliable in assessing hip adduction force by dynamometry, and clinical signs of athletes with groin pain. Hip adduction force seems an important clinical indicator as some studies have found hip adductor strength to be related to the incidence of groin injuries, (Crow, Pearce, Veale, Vanderwesthuizen, Coburn, & Pizzari, 2010; Fulcher et al., 2010) while others have shown that a strengthening program is of value in the treatment of chronic groin pain (Hölmich et al., 1999; Wollin & Lovell, 2006). Whilst these tests are showing clinical utility it would be valuable to ascertain if any clinical test positions are more specific to the different adductor muscles. This then may also provide direction for hip adductor strengthening rehabilitation programs.

Anatomically five muscles are likely to be involved in hip adduction and all have been implicated in various sports injuries. These muscles are pectineus, adductor brevis, adductor longus, gracilis and adductor magnus (Sinnatamby, 2003, pp. 113–118). Recent MRI and anatomical dissection (Robinson et al., 2007; Zoga et al., 2007) and reviews (Lovell, 2009; Robertson, Barker, Fahrer, & Schache, 2009) have also focused on the origins of these muscles

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around the pubic symphysis noting the existence of a common aponeurotic plate over the symphysis. No gait or athletic studies have examined the EMG activity of the individual hip adductors and there have been no studies on EMG activity in adductor clinical examination tests.

EMG has often been used to delineate between the individual muscle activation levels produced by various exercises (Ayotte, Stetts, & Keenan, 2007; Bolgla & Uhl, 2005; Bolgla & Uhl, 2007; Distefano, Blackburn, Padua, & Marshall, 2009; Earl, Schmitz, & Arnold, 2001). The disconnect between the measured EMG activity and the force produced by that muscle especially during movement has been long documented (Madeleine, Bajaj, Sogaard, & Arendt-Nielsen, 2001; Metral & Cassar, 1981; Perry & Bekey, 1981). Due to the relatively isometric nature of the hip adductor clinical tests, EMG was deemed an appropriate comparative measure for the contribution of the various adductor muscles. However, there are some methodological issues that need to be considered since the appropriate placement of electrodes for the hip adductor muscles remains unclear in the literature (see *Methods Section*).

The aims of this study are to determine the relative activation levels of the various hip adductor muscles in a variety of standard clinical tests in a group of athletes involved in a sport commonly associated with groin injuries, and to assess the hip adduction force produced in a subgroup of the tests. A further aim is to retrospectively examine the relationship between a previous history of groin injury and adductor activation. Clinically we aim to define the best tests to differentially activate the adductors.

2. Methods

2.1. Participants

Following institutional ethics approval athletes from an elite junior soccer program were invited to participate in this study. Athletes were excluded if they had a current lower limb injury, a fear of needles or a medical illness (eg recent illness, diabetes, bleeding disorder). Athletes' age, height, weight, dominant (kicking) leg and prior history of groin pain were recorded. A history of groin pain was regarded as positive if the athlete had previously missed 3 or more weeks of training and playing. This time frame was chosen as a clinical decision to indicate a significant groin injury. Athletes were excluded if they had groin pain at the time of the study or were not training fully for the previous 6 weeks.

2.1.1. Electrode placement

The identification of four adductor muscles (adductor longus, adductor magnus, pectineus and gracilis) was achieved using real-time ultrasound (5 MHz curved array and 10 MHz linear array transducers, Aloka SSD-900: Aloka Co. Ltd, Tokyo, Japan). Adductor brevis was considered too deep and therefore difficult to access for the purpose of this study. The landmarks that were identified were:

1. The anterior and posterior longitudinal edges of the gracilis
2. The longitudinal line of the fascial plane of the meeting of the adductor magnus and the medial hamstrings.
3. The medial edge of the sartorius.
4. The intersection of the sartorius and the gracilis.
5. The medial edge of the femoral neurovascular bundle in the inguinal region.
6. The inguinal ligament.
7. The superior anterior edge of the adductor longus in the inguinal region.
8. The intersection of the femoral neurovascular bundle and the adductor longus.

As noted earlier the placement of electrodes for the hip adductor muscles remains unclear in the literature. The general guidelines for electrode placement outlined by SENIAM were followed for the gracilis, adductor magnus and adductor longus of both legs (SENIAM, 2009). These muscles, however, are not included in their specific electrode placement recommendations. A point was selected approximately 50% between the distal motor endplate zone and the distal tendon away from the 'edge' of the muscle was utilised. The gracilis electrode was positioned longitudinally as mentioned above midway between the anterior and posterior edges previously identified on ultrasound. The adductor magnus electrode was similarly longitudinally positioned midway between the identified lines of the posterior edge of the gracilis and the longitudinal fascial plane between the adductor magnus and the medial hamstrings. The adductor longus electrode was positioned longitudinally as close to the above recommendations as possible; we placed the electrode between the anterior edge of the gracilis and the medial edge of the sartorius approximately 2 cm above the intersection of these two muscles (see Fig. 1).

The placement of the pectineus surface electrode is limited by the amount of surface area it has. We identified the small triangular surface area available, bound superiorly by the inguinal ligament, laterally by the femoral neurovascular bundle, medially by the adductor longus and inferiorly by the intersection of the previous two. A specialised electrode (see below) was placed centrally in this area. In addition a fine-wire electrode was inserted into the left pectineus just superiorly to the surface electrode. In the one study examining EMG activity of the pectineus fine-wire electrodes were used (Takebe, Vitti, & Basmajian, 1974).

A large indifferent electrode was placed on the proximal shaft of the left tibia.

2.1.2. Electrodes

Surface EMG recordings on the gracilis, adductor magnus and adductor longus were made using bipolar Ag/AgCl electrodes, of 10 mm diameter and fixed interelectrode distance of 20 mm (Viasys Neurocare, WI, USA). The surface electrodes used for the pectineus were custom made. They were pre-amplified bipolar stainless steel electrodes of 4 mm diameter and a fixed interelectrode distance of 12 mm. The fine-wire electrode was a commercially available stainless steel dual wire hook electrode, wire length 50 mm inserted by a 30 mm × 27 gauge needle (Viasys Neurocare, WI, USA).



Fig. 1. Patient starting position and EMG electrode placement. [patient consent has been obtained to use this image].

2.1.3. Skin preparation

Skin preparation was according to SENIAM guidelines (SENIAM, 2009). Where necessary areas were shaved, then lightly abraded with a gel abrasive (Nuprep, Weaver and Company, Co, USA). The areas were finally cleaned with an alcohol wipe. The left pectineal fine-wire electrode was placed before the surface electrode and was done under sterile conditions.

2.2. Testing procedure

After electrode placement (Fig. 1), athletes were transferred to a clinical couch where the electrodes were connected to a 16 channel Noraxon Myosystem 1400A (Noraxon, Scottsdale, USA) sampling at 1000 Hz. Adductor brevis was not sampled due to its difficult location under adductor longus. Before the clinical examination tests were performed a single hip adduction test was performed (corresponds to Hips 0 as outlined below which was considered the easiest and most familiar of the tests); this same test was also repeated after the completion of the clinical test array. This pre and post-test was used to examine the within test reliability of the EMG and force measures to ensure there was no change in signal due to electrode failure. An initial observation of the EMG signal was performed at this time and if possible problems with leads, electrode fixation or electrode connection were rectified.

2.3. Clinical tests

From the published literature on groin examination in athletes with groin pain, six tests were identified that would examine hip adduction and/or were stated to examine one muscle specifically. The tests identified were –

1. Position 'Hips 0': athlete supine, hips 0 deg flexion, knees 0 deg flexion, fist between medial femoral condyles. Test: resisted bilateral hip adduction (Fig. 2a) (Fulcher et al., 2010; Malliaras et al., 2009). This test was also used as the pre and post-test series check (see above)
2. Position 'Hips 45': athlete supine, hips 45 deg flexion (measured by inclinometer on the mid thigh (ISOMED, Portland, U.S.A.)), knees 90 deg flexion, feet flat on bed, fist between medial femoral condyles. Test: resisted bilateral hip adduction (Crow et al., 2010; Fulcher et al., 2010; Malliaras et al., 2009; Mens, Vleeming, Snijders, Ronchetti, & Stam, 1976) (Fig. 2b).
3. Position 'Hips 90': athlete supine, hips 90 deg flexion (measured by inclinometer on the mid thigh (ISOMED, Portland, U.S.A.)), knees 90 deg flexion (and supported by examiner), fist between medial femoral condyles. Test: resisted bilateral hip adduction (Fulcher et al., 2010; Rodriguez, Miguel, Lima, & Heinrichs, 2001) (Fig. 2c).



Fig. 2. a–f. Clinical examination tests displaying position (and with dynamometer position in 2 a,b,c and f). [patient consent has been obtained to use these images] a. Hips 0 (see text), b. Hips 45 (see text), c. Hips 90 (see text), d. Hips 0 ankle (see text), e. Hip 70 position (see text), f. Side lay position (see text).

4. Position 'Hips 0 ankle': athlete supine, hips 0 deg flexion, knees 0 deg flexion, elbows/palms between ankles, toes vertical. Test: resisted bilateral hip adduction (Hölmich et al., 2004) (Fig. 2d).
5. Position 'Hip 70': athlete supine, one hip 70 deg flexion (measured by inclinometer on the mid thigh (ISOMED, Portland, U.S.A.)), knee 100 deg flexion, stabilise opposite iliac crest. Test: resisted hip adduction of bent knee (towards opposite hip) [described as a specific test for pectineus (Winkel, 1997, pp. 40–41)] This was performed on both sides (Fig. 2e).
6. Position 'Side lay': athlete side-lying, legs straight and apart 20–30 deg, upper leg held by the examiner. Test: resisted hip adduction of the lower leg (Kendall & McCreary, 1983, pp. 176–177). This was performed on both sides (Fig. 2f).

For each of the tests an isometric hold with a 2 s build up and a 5 s hold was repeated three times. An internet research randomiser was used to randomise the order of tests (Research randomiser, 2009). A load cell (1 KN Xtran (Applied Measurements, Melbourne, Australia)) placed between the medial femoral condyles was used to provide an objective strength measure in tests 1,2,3 and 6. For tests 4 & 5 normal manual resistance was used. The load cell was connected to the Myosystem (1400A Noraxon, Scottsdale, USA) to gather the force data in real time with respect to the EMG data. Before each subject the load cell was calibrated using a 10 kg weight.

2.4. EMG data analysis

EMG recordings were analysed using Noraxon MRXP Master Edition Version 7.1 software (Noraxon, Scottsdale, Arizona, U.S.A.). The EMG signal was first visually inspected for major artefact and channels excluded if there appeared to be any failure in the test series. The EMG data was high pass filtered at 10 Hz, rectified and smoothed using 50 ms RMS (root mean square) moving window to create a contraction linear envelope. A moving average window of 1000 ms was then used to ascertain average peak activity during each of the tests. The highest moving average window for each muscle across the tests was delineated as 100% (this was delineated to be each subject and each muscle's maximum voluntary contraction (MVC)) and the subsequent tests' maximum 1000 ms moving window were normalised as a percentage of the MVC. For the single leg tests (Hip 70, Side lay) only the EMG data from the tested side was used in the analysis.

2.5. Statistical analysis

A mixed linear model with subject as a random effect and test order, test type, height, weight, body mass index (BMI), age, past injury history, side and dominant side as fixed effects was used to determine the factors influencing muscle activity and force production. Within subject paired *t*-tests were used to determine the within test reliability of electrode signal. Where appropriate effect sizes (ES) and 95% confidence intervals of the difference (95% CI Diff) are reported. In all data analysis statistical significance was assessed at the 95% level unless otherwise stated. Standard error of measurement was calculated as described by Hopkins (Hopkins, 2000).

3. Results

Twenty six athletes were invited to participate in the study. Five were excluded (two had lower limb injuries and three athletes declined due to needle aversion), leaving 21 athletes who completed the study. All were male players in an elite junior soccer program. The mean age of the athletes was 16.6 years (range

Table 1

This compares the MVC (maximum voluntary contraction) signal at the Hips 0 position of each muscle pre and post the testing procedure. For all measures besides the pectineus (fine-wire) there was no significant change, however there was wide variability.

Muscle	Post/Pre% Mean (SD)	<i>p</i> value	ICC
Gracilis	109.4 (35.3)	0.11	0.77
Add Mag	107.1 (34.4)	0.22	0.74
Add Long	109.3 (31.8)	0.1	0.77
Pec (surface)	97.6 (34.1)	0.67	0.66
Pec (fine-wire)	55 (21.3)	<0.01	0.47
Force	98.6 (15.1)	0.7	0.83

15.9–17.4), mean height was 178.5 cm (range 161–192) and mean weight was 73.1 kg (range 55–89). Fifteen athletes were right foot dominant (6 athletes were left foot dominant) and 6 athletes had a prior history of groin pain (4 athletes had osteitis pubis, 2 had hip adductor tendon injuries). All previous injuries were confirmed by review of documented medical history and MRI (magnetic resonance imaging) scans. All previously injured athletes had returned to full sporting activity for at least 8 weeks (Range 8–52, Mean 27.8, SD 14.8 weeks). There was no significant difference in the demographic data for the subjects with and without a groin injury.

3.1. Within test reliability of EMG signal

For the surface EMG signals only a subgroup of the adductor magnus readings were discarded. Seven series of tests of the adductor magnus were discarded due to either electrode displacement or cable failure during the testing procedure. All other surface EMG recordings were used in the analysis.

The checks on the stability of the EMG and force data measured from the pre- and post- experimental adductor squeeze test series can be seen in Table 1. For all measures of surface EMG and force the ICCs (intraclass correlation coefficients) ranged from moderate to good (0.66–0.83). The mean difference (mean diff) (Test 1- Test 2) for the surface EMG was very good ranging from only –2.1–5.1% MVC, however the SEM's ranging from 12 to 20% MVC suggest high variability. In the pectineus fine-wire EMG series there were five outright failures during the testing. Of the remaining 16 that still appeared on visual inspection to be producing viable EMG signal there was considerable decline of the signal on average of approximately 28.3% with low reliability (ICC = 0.48). For further analysis of the pectineus role in the various tests the fine-wire data was disregarded.

3.2. Test differences

A mixed linear model with subject as a random effect and test order, test type, height, weight, body mass index (BMI), age, past injury history, side, and dominant side as fixed effects was run for each muscle and the force data.

3.3. Force

The results for the force data can be seen in Fig. 3. Test type was a significant factor ($p < 0.01$) with Hips 0 being significantly stronger than Hips 45, Hips 90 and Side lay (ES 0.58, 1.35, 1.03; 95% CI Diff 0–9 kgs, 5–13 kgs, 3–13 kgs). Hips 45 was significantly stronger than Hips 90 (0.76; 1–9 kgs) but not Side lay. There was no significant difference between Hips 90 and Side lay. BMI was also a significant factor ($p < 0.01$) with players with a higher BMI producing a higher force. All other factors including injury history had no significant effect on the force outputs.

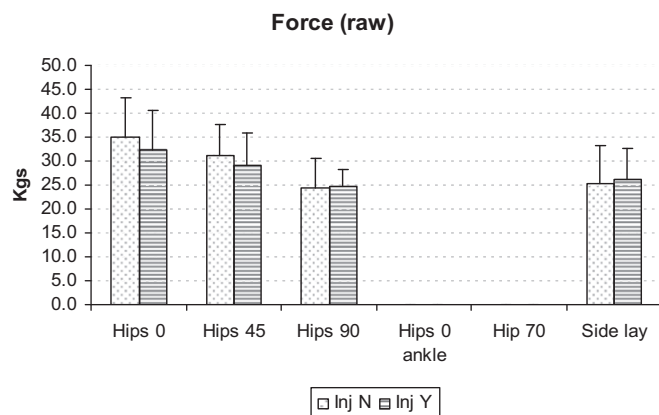


Fig. 3. Force data at each tested position.

3.4. Muscle EMG activity

There was no significant relationship of the EMG output of each of the muscles for side or for dominance in any of the test positions so left and right muscle data were collapsed. The pooled results for each muscle can be seen in Table 2.

Test type was a significant factor in the EMG output for all four muscles (all muscles $p < 0.01$). Injury history was a significant factor in the EMG output for the adductor longus ($p < 0.05$), pectineus ($p < 0.01$) and gracilis ($p < 0.01$) but not adductor magnus. All other factors appeared to have little relationship with the muscle EMG output. The separated data of those with and without a history of injury can be seen in Fig. 4.

The following results are describing the relative relationship between the 6 tests using the combined injured and non-injured data for each muscle. This is to identify the 'best' position or positions for testing each muscle, the magnitude of the differences of the tests (ES, 95% CIDiff) and the order of the test's level of activation produced for that muscle. Further comment on whether a history of injury affected the EMG output for that muscle is also included. For visualisation of the data see Fig. 4.

3.5. Adductor longus

The adductor longus had reasonably high EMG output for all the test positions. The combined data (injured and non-injured) demonstrated Hips 45 to have a statistically significant higher output than Hips 0, Hips 90, Hips 0 ankle and Hip 70 (ES- 0.83, 0.86, 0.61, 1.06; 95%CI Diff 7–21%, 7–22%, 3–19%, 11–27%) but not Side lay. Hips 0 ankle (0.39; 0–16%) and Side lay (0.58; 3–18%) were also significantly greater than Hip 70 (Order- Hips 45 > Side lay $\geq$ Hips 0 ankle $\approx$ Hips 0 $\approx$ Hips 90 $\geq$ Hip 70). A past history of injury was related to significant ($p < 0.05$) reduction of the EMG output of the adductor longus in Hips 90 (0.97; 5–27%), Hips 0 ankle (1.06; 7–33%) and Hip 70 (0.43; 0–20%).

Table 2

The mean (standard deviation) activation level of each muscle in each test position as a percentage of the individual muscle's MVC.

	Adductor longus	Adductor magnus	Gracilis	Pectineus
Hips 0	73.4 (17.4)	74.2 (20.4)	56.0 (19.0)	54.8 (20.1)
Hips 45	87.3 (15.9)	78.1 (25.9)	87.2 (17.5)	80.6 (17.8)
Hips 90	72.9 (17.7)	49.6 (23.4)	70.5 (21.5)	84.1 (19.7)
Hips 0 ankle	76.1 (20.4)	62.2 (24.1)	62.3 (22.0)	66.0 (23.6)
Hip 70	68.3 (19.7)	64.4 (24.4)	78.3 (18.0)	77.5 (19.7)
Side lay	78.6 (15.4)	47.5 (17.5)	68.1 (21.1)	69.2 (18.9)

3.6. Adductor magnus

The adductor magnus had a more selective output in the various test positions. The combined data demonstrated Hips 0 and Hips 45 (ES range 0.44–1.38; 95%CI Diff range 0–20%, 21–40%) had significantly greater EMG output than the other tests whilst Hips 90 and Side lay had significantly less than the other tests. (Order- Hips 45 $\geq$ Hips 0 > Hips 0 ankle $\approx$ Hip 70 > Side lay $\approx$ Hips 90). There was no significant difference in adductor magnus activation between those with and without a past history of injury although it appears that subjects with a past history of injury may predominately have their highest activation in Hips 45.

3.7. Gracilis

The gracilis demonstrated greatest activation in Hips 45 which was statistically greater than all other tests (ES range 0.5–1.7; 95%CI Diff range 1–16%, 23–39%). Hip 70 was the next most active test which was statistically greater than Hips 90 and Side lay. The second least active was Hips 0 ankle and Hips 0 was significantly less active than all other tests. (Order- Hips 45 > Hips 70 > Hips 90 = Side lay > Hips 0 ankle > Hips 0). A history of injury was represented by a significant ($p < 0.05$) decline in the EMG output in the Hips 0 (1.2; 9–32%) and Hips 0 ankle tests (1.08; 8–35%).

3.8. Pectineus

Pectineus had its greatest activation levels in the Hips 90 position. Hips 45, Hips 90 and Hips 70 are equivalent but Hips 90 was statistically greater than all the other tests (ES range 0.77–1.47; 95%CI Diff range 6–23%, 21–38%) whereas Hips 45 is only statistically greater than Hips 0 and Hips 0 ankle. Hips 0 ankle is also statistically less than Hips 70 (Order- Hips 90 $\geq$ Hips 45 $\geq$ Hip 70 > Side lay > Hips 0 ankle > Hips 0). A history of injury saw a statistically significant ($p < 0.05$) reduction in the EMG output in the Hips 0 (1.2; 9–34%) and Hips 0 ankle tests (1.36; 15–45%) compared with the other tests.

4. Discussion

The clinical diagnosis and ongoing assessment of athletes with groin pain has been a focus of recent research. More precise anatomical description of the musculature around the pubic symphysis, hip adduction dynamometry and investigation using better quality MRI are all assisting the practitioner in making a diagnosis in athletes with groin pain.

In this study EMG data was analysed as a percentage of a muscle's maximal output in this series of tests. As such this analysis only allows comparison of individual tests for each muscle and does not permit interpretation of all muscles comparatively at each clinical test. The high variability commonly associated with EMG studies and also demonstrated in this study must be considered as a limitation of the data. The factors included in the mixed linear model were all considered possible factors affecting EMG output. However, the relatively small number (21) of participants is a concern for the sophistication of the model used. For this reason we have presented the data in a raw form (see Fig. 4) and presented effect sizes and the 95% confidence interval of the differences demonstrated.

Clinical test type was a significant factor in the EMG output for each muscle. Using this approach to achieve maximal contraction of a muscle (after history and palpation assessment has suggested a likely source of pain or injury) the findings from this study suggest that the clinician should assess the gracilis at Hips 45 deg,

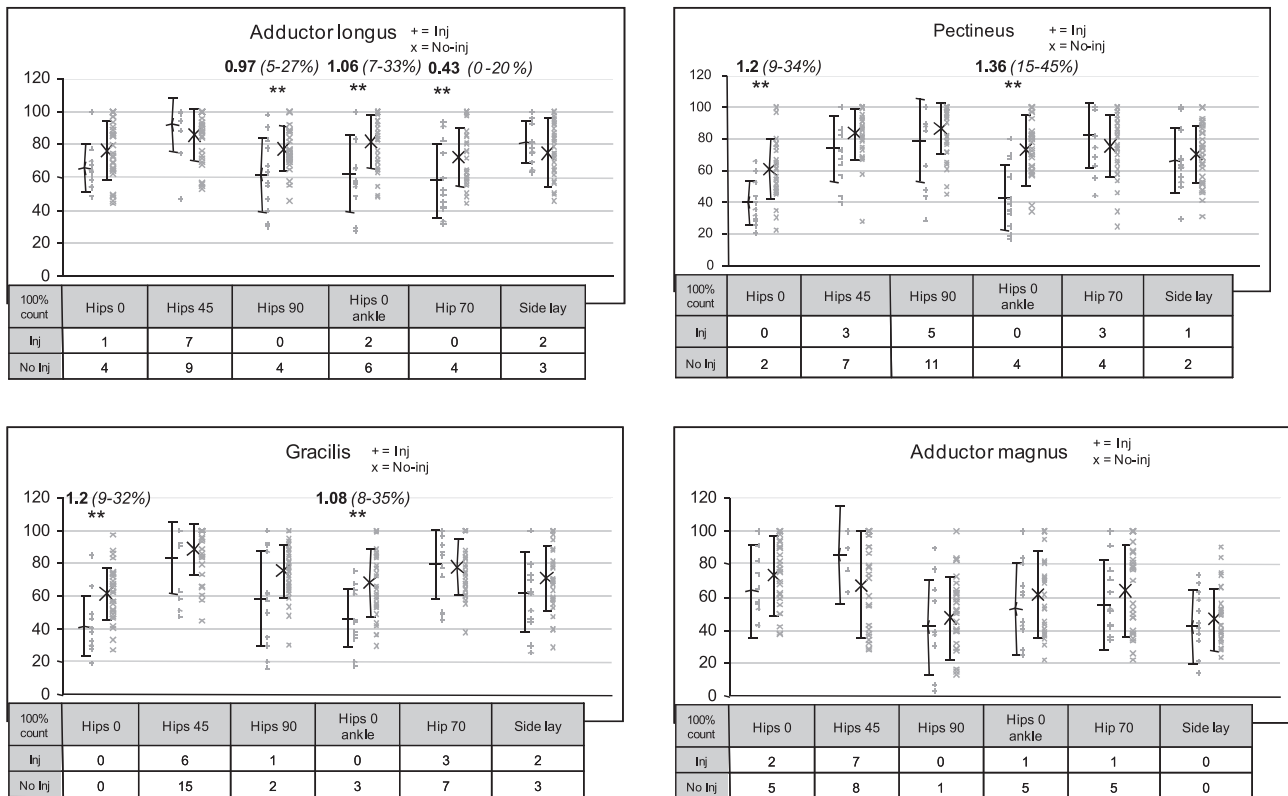


Fig. 4. The individual data points and the distributions of those with a past history of injury (Inj) and those without a past history of injury (No inj) are demonstrated here. As it is difficult to fully represent the number of times a particular test was the 100% benchmark for each muscle they are displayed in the table under each graph (** denotes a difference between Inj and No inj. $p < 0.05$, Effect size (95% confidence interval of the difference)).

adductor longus at Hips 45 deg, adductor magnus at Hips 45 or 0, and pectineus at Hips 90.

The separate muscular anatomical locations have always suggested diverse roles in hip adduction during gait and this is supported by the EMG signals in the various testing positions. It does seem plausible that the pectineus, due to its hip flexor role, would be more active in the Hips 90 as it may be contributing to the hip flexion. Similarly, the higher activation of adductor magnus in Hips 0 may be due to its relationship with the hamstrings. It is interesting to note that while the force output at Hips 0 was the highest this was not the position of maximal muscle activation of each adductor. Other muscles not tested in this study such as the hamstrings may be contributing to hip adduction in this position also due to its relative size the adductor magnus may contribute more to the force output in this position. Further testing of the individual adductor muscles during gait and sporting activities may provide useful information for clinicians managing groin injuries.

Larger athletes are likely to produce higher forces and this study has shown a positive correlation with BMI. In their study of normative data on hip flexion and adduction using hand held dynamometry, Hanna, Fulcher, Elley, and Moyes (2010), demonstrated a correlation with dominant side and weight for hip flexion and for height with hip adduction. Recently, good reliability has been found using hand held dynamometry testing of hip adduction by several authors (Fulcher et al. (2010); Malliaras et al., 2009; Thorborg, Petersen, Magnusson, & Hölmich, 2010) and by manual strength testing by one author (Hölmich et al., 2004). The positions found reliable in these studies have been Hips 0, 45, 90 and resisted Hips 0 ankle. One study found the Hips 90 position was more reliable than the other positions and suggested that greater strength may adversely affect reliability using hand held dynamometry (Fulcher et al., 2010). While our force data confirm lower

force values are found in this position (Hips 90), it should be noted that there is different muscle activation (pectineus higher activation, adductor magnus lower activation) which should therefore be interpreted in a different clinical manner. Our force data suggest that overall strength can best be assessed in either the Hips 0 or 45 positions, supporting current clinical assessment positions. Further, when combining the force and EMG data, these specific clinical examination test positions may be appropriate to use in isometric hip adductor exercise programs in the early stages of rehabilitation of groin injuries.

A past history of groin injury is related to a significant fall in EMG signal in Hips 90, Hips 0 ankle and Hip 70 positions whether this is a precipitating or subsequent factor is unclear and beyond the scope of this project. Current research has identified a rectus abdominis/adductor aponeurotic (RA-AA) plate overlying the pubic symphysis with MRI evidence suggesting injury to this structure and/or the pubic symphysis is likely to be the cause of many groin injuries in athletes (Lovell, 2009; Robertson et al., 2009). The adductor longus is the primary adductor muscle that originates from this aponeurosis and, assessment of this muscle may be a primary focus of clinical examination and rehabilitation. EMG and force data in athletes with a past history of groin pain would suggest that adductor longus strength needs to be assessed in multiple clinical positions. It was also noted that in this group EMG activity at the Hips 0 ankle position declined significantly. Examination in this position may be clinically important and a similar position of single leg hip adduction against a hand-held dynamometer at the ankle has recently been found to be a reliable measure in healthy volunteers (Thorborg et al., 2010). Our study suggests that these athletes have not returned to a 'normal' state which may be due to inadequate rehabilitation or to a permanent alteration to the RA-AA plate with subsequent changes in

biomechanics at the pubic symphysis. Either or both could explain the frequent recurrence of athletic groin pain.

For the surface EMG tests and the force data the within test reliability was good and athletes were able to replicate their pre-test scores post-test. The decline in pectineus fine-wire EMG recordings seen in this study was probably due to wire movement during the testing procedure.

5. Conclusion

Previous clinical commentaries have not considered the specificity of adductor muscle activation in different test positions. Whilst active in most test positions, pectineus is best assessed at Hips 90 whilst the other adductors can be assessed in Hips 45 or Hips 0. Hips 0 had the highest force output and Hips 45 generally the highest EMG output. Athletes with a previous groin injury had significant differences in hip adductor muscle activation of adductor longus, pectineus and gracilis compared with uninjured athletes. These findings have major implications for the understanding of groin injuries particularly in regard to rehabilitation of athletes, new anatomical dissection work and for the direction of further investigation. Current examination positions used in handheld dynamometry studies are appropriate for assessing the hip adductor muscles.

Conflict of interest

No conflict of interest.

Ethical approval

Australian Institute of Sport Ethics Committee, approval number: 20090201.

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